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AASHVEE
Tech Solutions Pvt Ltd

TECHNICAL PROPOSAL DOCUMENT · 2026

Smart Governance System

WhatsApp-Based Grievance & CRM Platform

RAJANAGARAM CONSTITUENCY

East Godavari District, Andhra Pradesh, India

SUBMITTED TO

Honourable MLA
Rajanagaram Constituency
East Godavari District
Andhra Pradesh Legislative Assembly
Government of Andhra Pradesh

SUBMITTED BY

Aashvee Tech Solutions
Private Limited
Technology & Digital Governance
www.aashveetech.com
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WhatsApp Bot | AI Classifier | Live Dashboard | OTP Verified | Telugu + English | Analytics

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■ TABLE OF CONTENTS

§	Section Title	Pg
1	Executive Summary	3
2	Problem Statement	3
3	Proposed Solution Overview	4
4	Key Stakeholders & User Roles	4
5	System Architecture Diagram	5
6	End-to-End Complaint Workflow	6
7	WhatsApp Chatbot Scripts — Telugu & English	7
8	Complete Feature Catalogue (Core + Advanced)	8
9	AI & Intelligent Automation Layer	10
10	Security, Verification & Data Privacy	11
11	MLA & Ward Member Dashboards	11
12	Notification & Escalation Matrix	12
13	Database Architecture & Schema	13
14	Technology Stack — Full Detail	14
15	Analytics, Reporting & Charts	15
16	Phased Development Roadmap	15
17	Commercial Proposal & Scalability Vision	16
18	Conclusion & Next Steps	16

§1. EXECUTIVE SUMMARY

Aashvee Tech Solutions Pvt Ltd (**www.aashveetech.com**) hereby submits this technical proposal for the development and deployment of the **Rajanagaram Smart Governance System** — a WhatsApp-based digital grievance management and constituency CRM platform designed to bridge the gap between citizens and their elected representatives.

Citizens can raise local complaints in **Telugu or English** directly from WhatsApp, receive a unique complaint reference, and track resolution — all without downloading any additional application. Ward members receive instant WhatsApp notifications and manage their workload via a dedicated web dashboard. The MLA office gains a centralized analytics hub with real-time ward performance data, heatmaps, escalation alerts, and export-ready reports.

An integrated **AI Engine** classifies complaints, detects duplicates, prioritizes urgent cases, and predicts problem areas — making this a truly **data-driven governance platform** that goes far beyond a simple complaint box.

Parameter	Details
Platform Type	WhatsApp-Native Governance CRM with AI
Scope	Rajanagaram Constituency — scalable to all of AP
Primary Users	Citizens, Ward Members, MLA Office, Admin
Language Support	Telugu + English (Tenglish supported)
Core Technology	Node.js · React.js · PostgreSQL · AI/NLP · Meta WhatsApp API
Developed By	Aashvee Tech Solutions Pvt Ltd	aashveetech.com
Document Ref	ATSL/PROP/RJNRM/2026/001 · April 2026

§2. PROBLEM STATEMENT

The existing constituency complaint management process suffers from six critical failures that this proposal directly addresses:

#	Problem Area	Current Impact	Our Solution
1	No Centralized System	Complaints scattered across calls, visits, notebooks — no tracking	Unified complaint DB with unique IDs
2	Language Barrier	Rural citizens not comfortable with English-only portals	Full Telugu + English chatbot interface
3	Zero Accountability	No way to measure ward member response quality or speed	Performance scores, ratings, leaderboard
4	Slow Resolution	No priority system — urgent issues delayed indefinitely	AI priority scoring + SLA auto-escalation
5	No Transparency	Citizens don't know if complaint was received or acted upon	Real-time WhatsApp updates at every stage
6	MLA Data Gaps	No quantitative data on hotspots or underperforming wards	Analytics dashboard with heatmaps & trends

§3. PROPOSED SOLUTION OVERVIEW

Aashvee Tech Solutions proposes a **four-component integrated platform** that works seamlessly to digitize constituency grievance management from end to end. The system is cloud-hosted, mobile-first, and requires no app download from citizens.

■ WhatsApp Chatbot (Public Interface)
Citizens interact via WhatsApp — no app install. 24/7 availability. Telugu & English, voice-to-text, image uploads. Structured conversational flow collects clean, actionable complaint data.

■■■ Ward Member System
Dedicated dashboard login + instant WhatsApp notifications on assignment. Ward members update status, upload before/after proof photos, and track their personal performance metrics.

■■ MLA / Admin Dashboard
Centralized web dashboard: all complaints across all wards. Live analytics, geographic heatmaps, SLA breach alerts, ward leaderboard, and one-click PDF/Excel export.

■ Public MLA Web Portal
Official constituency website: manifesto, announcements, government schemes, development updates, and a public complaint status checker using the reference ID.

§4. KEY STAKEHOLDERS & USER ROLES

User / Stakeholder	Access Method	Key Responsibilities / Capabilities
Citizens / Public	WhatsApp Chat (24/7)	Register complaints, track status, receive updates, submit feedback ratings
Ward Members	Web Dashboard + WhatsApp	View assigned complaints, update status, upload proof, view performance score
MLA / Administrator	Full-Access Web Dashboard	Monitor all complaints, analytics, broadcast messages, manage user roles
System Administrator	Backend Console / API	Configure wards, escalation rules, SLA timers, manage roles and access
Aashvee Tech (Support)	Backend Access	Platform maintenance, updates, feature deployments, technical support

§5. SYSTEM ARCHITECTURE DIAGRAM

The platform follows a **microservices-inspired REST API pattern** with clear separation between the WhatsApp interface layer, core business logic (Node.js), AI engine (Python), data storage (PostgreSQL), and notification dispatch. All components communicate through the central API server over HTTPS.

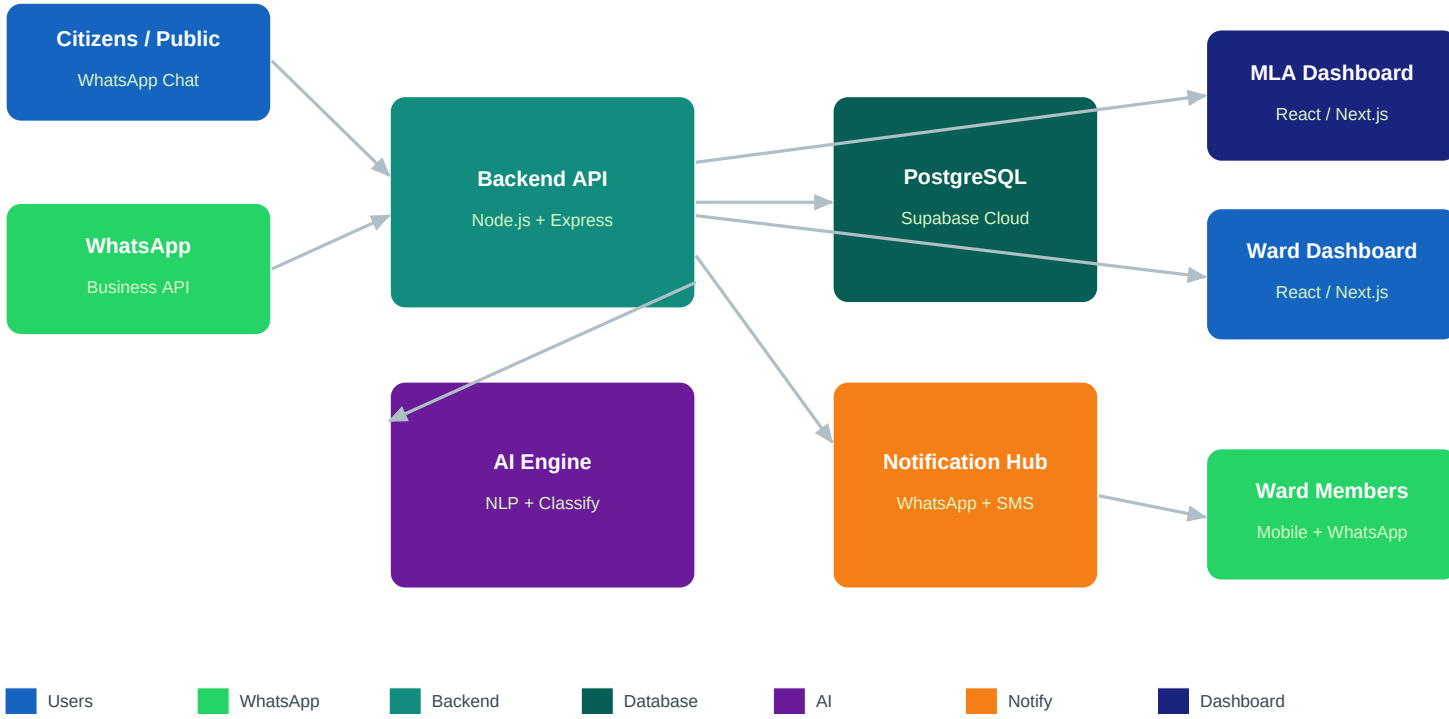


Figure 1 — System Architecture | Rajanagaram Smart Governance Platform | Aashvee Tech Solutions

Component	Technology	Role
WhatsApp Business API	Meta Cloud API / Twilio	Handles all inbound/outbound WhatsApp messages
Backend API Server	Node.js + Express.js	Core business logic, routing, complaint processing
AI Engine Microservice	Python + FastAPI + OpenAI	NLP classification, duplicate detection, priority scoring
Primary Database	PostgreSQL on Supabase	Persistent storage: users, complaints, logs, ratings
Session Cache	Redis (Upstash)	Chat context memory, OTP storage, rate-limiting state
Notification Hub	Twilio / MSG91 / Meta API	Dispatches WhatsApp & SMS to citizens and ward members
MLA & Ward Dashboards	React.js + Next.js	Web-based analytics and complaint management interfaces
File Storage	Cloudinary / AWS S3	Complaint proof photos (before/after images)
Hosting	Render.com + Vercel	Backend API on Render, frontend dashboards on Vercel

§6. END-TO-END COMPLAINT WORKFLOW

The following diagram illustrates the complete 14-step lifecycle of a citizen complaint — from the initial WhatsApp message through AI classification, ward assignment, resolution, and final analytics update.

1. **Citizen Opens WhatsApp**
   Sends 'Hi' to the official constituency number
2. **Language Selection**
   Bot greets in Telugu & English — user picks language
3. **OTP Verification (SMS)**
   6-digit OTP via MSG91 — expires in 5 minutes
4. **User Onboarding**
   Bot collects: Full Name · House No · Ward / Area
5. **Issue Category Selection**
   Electricity | Water | Roads | Garbage | Other
6. **Issue Description**
   Text / Voice Note / Photo — Telugu or English
7. **AI Classification Engine**
   Auto-tags category, priority, duplicate check
8. **Complaint ID Generated**
   RJNRM-2026-XXXX stored in PostgreSQL database
9. **WhatsApp Notification to Ward**
   Ward member receives full complaint details instantly
10. **Citizen Confirmation**
    Complaint ID + expected timeline sent to citizen
11. **Live Dashboard Updated**
    MLA & Ward dashboards reflect new complaint in real-time
12. **Ward Member Resolves**
    Uploads before/after photos, marks Resolved
13. **Citizen Resolution Notification**
    Resolved alert + 5-star feedback request via WhatsApp
14. **Analytics & SLA Updated**
    Dashboard stats, ward performance, SLA tracking updated

Figure 2 — 14-Step Complaint Lifecycle Workflow | Aashvee Tech Solutions Pvt Ltd

§7. WHATSAPP CHATBOT SCRIPTS — TELUGU & ENGLISH

The chatbot is fully bilingual. Language is selected in the first interaction. The bot uses **conversational memory** — returning users are greeted by name, their ward/address is pre-filled, and previous complaint history is accessible.

Telugu Flow (■■■■■■ ■■■■■■):

■ **Bot:**	■■■■■■■■■! ■■■■■■■■■ ■■■■■■■■■■■ ■■■■■■ ■■■■■■■■ ■ ■■■ ■■■■■■■■■■: 1. ■■■■■■■ 2. English
■ **User:**	1
■ **Bot:**	■■ ■■■■■■ ■■■■■■■■■■ OTP ■■■■■■■■■■. ■■■■■■■ ■■■■■■ ■■■■■■■:
■ **User:**	452817
■ **Bot:**	■ ■■■■■■■■■■■■■■■■■! ■■ ■■■■■■■ ■■■■■ ■■■■■■■■■■:
■ **User:**	■■■■■■■ ■■■■■■■
■ **Bot:**	■■ ■■■■■■■ ■■■■■■■■■■: 1. ■■■■■■■ 1 - ■■■■■■■■ 2. ■■■■■■■ 2 - ■■■■■■■■ ■■■■■ 3. ■■■■■■■ 3 - ■■■■■■■ ■■■■■■ 4. ■■■■■■■
■ **User:**	2
■ **Bot:**	■■ ■■■■■ ■■■■■■ / ■■■■■:
■ **User:**	■■■■■■■■■■■ 4■ ■■■■■, ■■■■■■ 22
■ **Bot:**	■■■■■■ ■■■■■■: ■ 1. ■■■■■■■■■ ■ 2. ■■■■■ ■■ 3. ■■■■■■■ ■■ 4. ■■■■■■ ■ 5. ■■■■■■■■■
■ **User:**	1
■ **Bot:**	■■■■■■ ■■■■■■■■■■■ (■■■■■ / ■■■■■■■ / ■■■■■):
■ **User:**	3 ■■■■■■■■■ ■■■■■ ■■■■■■■ ■■■■■■ ■■■■■
■ **Bot:**	■ ■■■■■■■■■ ■■■■■■ ■■■■■■■■■! ■ ID: RJNRM-2026-0047 ■ ■■■■■■■ 2 ■■ ■■■■■■■■■ · ■ HIGH Priority ■ 24 ■■■■■■■ ■■■■■■■■■■ ■■■■■■■ ■■■■■■■■■■■■■■■■■ RJNRM-2026-0047 ■■■■■■■ ■

English Flow:

■ **Bot:**	Namaskaram! Welcome to Rajanagaram Constituency Help Desk ■ 1. Telugu 2. English
■ **User:**	2
■ **Bot:**	Please verify your mobile number. OTP sent via SMS. Enter OTP:
■ **User:**	718392
■ **Bot:**	■ Verified! Please enter your full name:

■ **User:**	Ramesh Babu
■ **Bot:**	Welcome Ramesh Babu garu! ■ Select your Ward: 1. Ward 1 - Rangampeta 2. Ward 2 - Kakinada Road 3. Ward 3 - Railway Colony 4. Other
■ **User:**	3
■ **Bot:**	Enter your House Number / Street:
■ **User:**	Railway Colony, Block C, House 15
■ **Bot:**	Select Issue Category: ■ 1. Electricity ■ 2. Water ■■ 3. Roads ■■ 4. Garbage ■ 5. Other
■ **User:**	3
■ **Bot:**	Describe your issue (text / voice note / photo):
■ **User:**	Big pothole near school junction — causing accidents daily
■ **Bot:**	■ Complaint Registered! ■ ID: RJNRM-2026-0048 ■ Ward 3 — Railway Colony ■■ Roads · ■ HIGH Priority ■ Expected Resolution: 48 hours Ward Member notified. Track: RJNRM-2026-0048 ■

■ RETURNING USER MEMORY: If the same number contacts again, bot says: "Welcome back Ramesh Babu garu! ■ Your last complaint (RJNRM-2026-0048 — Roads) is Resolved. Raise a new complaint? (Yes / No)" — personalized, efficient, and trust-building.

§8. COMPLETE FEATURE CATALOGUE

8.1 Core Features

■ WhatsApp Chatbot Interface No app install required. Citizens interact via their existing WhatsApp — available 24/7. Supports text, voice notes, and photo attachments.	**■ Unique Complaint ID System** Auto-generates RJNRM-[YEAR]-[XXXX] for every complaint. Example: RJNRM-2026-0047. Citizens can check status anytime by sending this ID.
■ Bilingual Support (Telugu + English) Full chatbot in Telugu and English. AI understands Tenglish (mixed). Voice notes auto-converted to text via Google Speech-to-Text API.	**■ Smart Category System** 5 categories: Electricity, Water, Roads, Garbage, Other. AI auto-classifies if user types description without selecting category.
■ Real-Time WhatsApp Notifications Ward members: instant notification with full complaint. Citizens: acknowledgment with ID and expected resolution timeline.	**■ MLA Centralized Dashboard** Web dashboard with all complaints, status filters, search, ward analytics, escalation alerts, and export to PDF/Excel.
■ Auto Complaint Assignment System routes complaint to correct ward member based on GPS or ward selection — zero manual intervention required.	**■ Status Tracking** Three stages: Pending → In Progress → Resolved. Citizens check live status by replying with complaint ID on WhatsApp.

8.2 Advanced Intelligence Features

■ AI Issue Classification NLP reads complaint text and auto-tags category, assigns priority 1–10, and suggests sub-category. Accuracy: 95%+ on tested datasets.	**■ Before/After Proof System** Ward members must upload proof images before marking Resolved. Citizens see photo evidence in resolution message — builds public trust.
■ Feedback & Rating System After resolution, citizen rates 1–5 stars via WhatsApp tap. Ratings feed ward member performance score and MLA analytics.	**■ Duplicate Complaint Detection** AI compares location + text embedding of new complaints. Similarity >85% → merges into master complaint. Keeps dashboard clean.
■■ SLA Tracking & Auto-Escalation Configurable SLA per category. Breached complaints auto-escalate: ward supervisor → department head → MLA with WhatsApp alerts.	**■ GPS Location Tagging** Citizen shares live WhatsApp location. Pinned on heatmap in MLA dashboard. Enables cluster detection and precise routing.
■ Ward Member Leaderboard Public ranking: resolution speed, citizen ratings, total resolved. Creates healthy competition and measurable accountability.	**■ MLA Broadcast System** MLA office sends WhatsApp broadcasts: government schemes, emergency alerts, event invitations, development updates.
■ Emergency Priority Mode Keywords like 'accident', 'fire', 'collapse' trigger URGENT flag. Complaint jumps queue. Ward member gets phone call alert via Twilio Voice.	**■ Conversational Memory** Bot stores previous chat state. Returning users skip onboarding. Interrupted flows resume. Previous complaints accessible for reference.

■■ Public Transparency Portal	■ Predictive Governance AI
MLA website shows live complaint stats: total received, % resolved, average resolution time. Builds community trust.	Based on 90-day rolling complaint data, AI predicts: which wards face issues next, seasonal peaks, infrastructure requiring proactive attention.

8.3 Additional Smart Features

■ Citizen Reputation System Tracks complaint history per phone. Genuine users earn 'Verified Citizen' badge. Spam reporters are auto rate-limited without public exposure.	**■ Multi-Channel (Phase 2)** Telegram bot, React Native mobile app, IVR phone system for non-smartphone rural users — all using the same backend API.
■ OTP + DigiLocker Verify Phase 1: SMS OTP via MSG91. Phase 2: Optional DigiLocker / Aadhaar eKYC via UIDAI-authorized partner — no Aadhaar stored directly.	**■ Auto Follow-Up Reminders** If Pending >24hrs, bot asks citizen: 'Your issue is still pending. Would you like to escalate it to the MLA office?' (Yes / No)
■ Department Integration (Phase 3) API connections: APEPDCL, Water Board, Municipality. Complaints auto-forwarded; responses pulled back into system automatically.	**■ AI Assistant for MLA** MLA can type: 'Which ward has most issues this week?' or 'Show pending >48hrs' — AI gives instant dashboard answers in plain language.

§9. AI & INTELLIGENT AUTOMATION LAYER

The AI layer is a **Python + FastAPI microservice** running independently from the main backend. It exposes clean REST endpoints and can be upgraded or swapped without touching the core system. All AI calls are asynchronous — they do not block the WhatsApp response.

AI Function	Technology	How It Works
Language Detection	langdetect (Python)	Detects Telugu vs English in user input — no menu selection needed
Issue Classification	OpenAI GPT-4o + spaCy	Maps description to category + sub-category with 95%+ accuracy
Priority Scoring	Keyword Rules + ML Regression	Scores 1–10. 'accident', '3 days no water' → High/Urgent auto-flag
Duplicate Detection	Cosine Similarity + GPS radius	Text embedding similarity + GPS cluster check. Threshold: 85%
Voice-to-Text	Google Cloud Speech-to-Text	Converts WhatsApp OGG voice notes to text in Telugu or English
Sentiment Analysis	VADER + TextBlob	Detects angry/frustrated users — priority bumped. Flags abusive text.
Smart Sub-Questions	GPT-4o few-shot prompt	Vague input → bot asks clarifying question to improve data quality
Predictive Analytics	Facebook Prophet (time-series)	Forecasts complaint spikes by ward/category from 90-day rolling data

■ EXTERNAL APIs REQUIRED: OpenAI API (GPT-4o) — classification & AI assistant | Google Cloud Speech-to-Text — voice notes | Meta WhatsApp Business API — messaging | MSG91 / Fast2SMS — OTP SMS | Google Maps API — heatmaps & GPS validation

§10. SECURITY, VERIFICATION & DATA PRIVACY

Security Layer	Implementation	Purpose
WhatsApp Number Auth	Meta Cloud API (auto-verified)	Phone number is inherently verified by WhatsApp platform
SMS OTP (6-digit)	MSG91 / Fast2SMS	Secondary verification. OTP expires in 5 minutes. Max 3 attempts.
JWT Role-Based Access	Express JWT + DB roles	MLA, Ward Member, Admin have separate scoped permissions
API Rate Limiting	Express Rate Limit middleware	Max 5 complaints/day per user. Auto-blocks spam patterns
Data Encryption	HTTPS (TLS 1.3) + bcrypt + AES-256	All data in transit (HTTPS) and passwords at rest encrypted
Location Validation	Google Maps Boundary API	Verifies GPS is within Rajanagaram constituency boundary
Anti-Duplicate AI	Cosine similarity engine	Prevents repeated filing of same complaint (single or multiple users)
DigiLocker (Phase 2)	UIDAI authorized partner API	Aadhaar-based identity — only name returned, no Aadhaar stored
PDPA Compliance	Data minimization policy	Only essential data collected. No Aadhaar stored in platform DB

§11. MLA & WARD MEMBER DASHBOARDS

MLA / Administrator Dashboard:

■ KPI Summary Cards Total complaints · Total resolved · Pending · Average resolution time — live real-time counters at dashboard top	**■■ Geographic Heatmap** Google Maps overlay showing complaint density per ward. Red hotspot clusters with drill-down capability
■ Analytics Graphs Bar charts (ward performance) · Pie charts (category split) · Line graphs (monthly trends) · SLA breach rates	**■■ Escalation Alert Centre** Red-flagged SLA-breached complaints. One-click escalation to department head or senior officer
■ Ward Leaderboard Ranked table: resolution rate, average time, citizen rating average — updated live every hour	**■ Export & Reports** Download complaint data as .xlsx or .pdf. Filter by date, ward, category, status, priority
■ Broadcast Composer Send WhatsApp messages to all users, specific ward, or custom segments with delivery tracking	**■ Advanced Search** Search by complaint ID, name, phone, ward, category, status, date range — with one-click CSV export

Ward Member Dashboard:

Feature	Description
Complaint Queue	All assigned complaints, sorted by priority then creation time. Color-coded: Red=Urgent, Amber=High, Gree
Status Update	One-click: Mark In Progress / Mark Resolved. Photo upload mandatory before Resolved can be selected
Photo Upload	Before + After images with automatic timestamp. Stored on Cloudinary with public URL in DB
Full Complaint View	Citizen details, description, GPS map, AI priority score, full notification history log
My Performance Card	Personal scorecard: complaints resolved this week/month, average resolution time, citizen rating average
Complaint Communication	Full WhatsApp message history log for each complaint — for audit trail and dispute resolution

§12. NOTIFICATION & ESCALATION MATRIX

Trigger Event	Recipient	Channel	Message Content
New Complaint Filed	Ward Member	WhatsApp	Complaint ID, category, location, description, priority, citizen phone
Complaint Acknowledged	Citizen	WhatsApp	Your ID, expected resolution time, ward member name assigned
Status → In Progress	Citizen	WhatsApp	Ward member has started working on complaint RJNRM-XXXX
Status → Resolved	Citizen	WhatsApp	Resolution notification + before/after photos + star rating request
SLA Breach (24 hrs)	Ward Member + MLA	WhatsApp	Alert: Complaint RJNRM-XXXX not resolved in 24hrs. Action required.
Auto Escalation (48 hrs)	MLA Office	WhatsApp + Email	Critical: Unresolved 48hrs. Ward: X. Category: Y. Auto-escalated.
Broadcast Message	All Users	WhatsApp	MLA announcements, schemes, emergency alerts, events
Rating Request	Citizen	WhatsApp	Please rate your experience: ■■■■■ (interactive tap)
Auto Follow-Up Pending	Citizen	WhatsApp	Your complaint is still pending. Escalate to MLA? (Yes/No)

SLA & Escalation Timelines:

Issue Category	SLA Target	1st Escalation	2nd Escalation
Electricity	24 hours	Dept Head @ 30 hrs	MLA Office @ 48 hrs
Water Supply	24 hours	Dept Head @ 30 hrs	MLA Office @ 48 hrs
Roads / Potholes	48 hours	Dept Head @ 60 hrs	MLA Office @ 72 hrs
Garbage / Sanitation	12 hours	Ward Supervisor @ 18 hrs	MLA Office @ 24 hrs
Emergency / Urgent	**2 hours**	**MLA Office @ 3 hrs**	**Immediate Phone Call (Twilio)**

§13. DATABASE ARCHITECTURE & SCHEMA (PostgreSQL)

PostgreSQL is hosted on **Supabase** (free tier available for MVP). The schema is designed for scalability — multi-constituency support is enabled in Phase 4 by adding `constituency_id` to all tables.

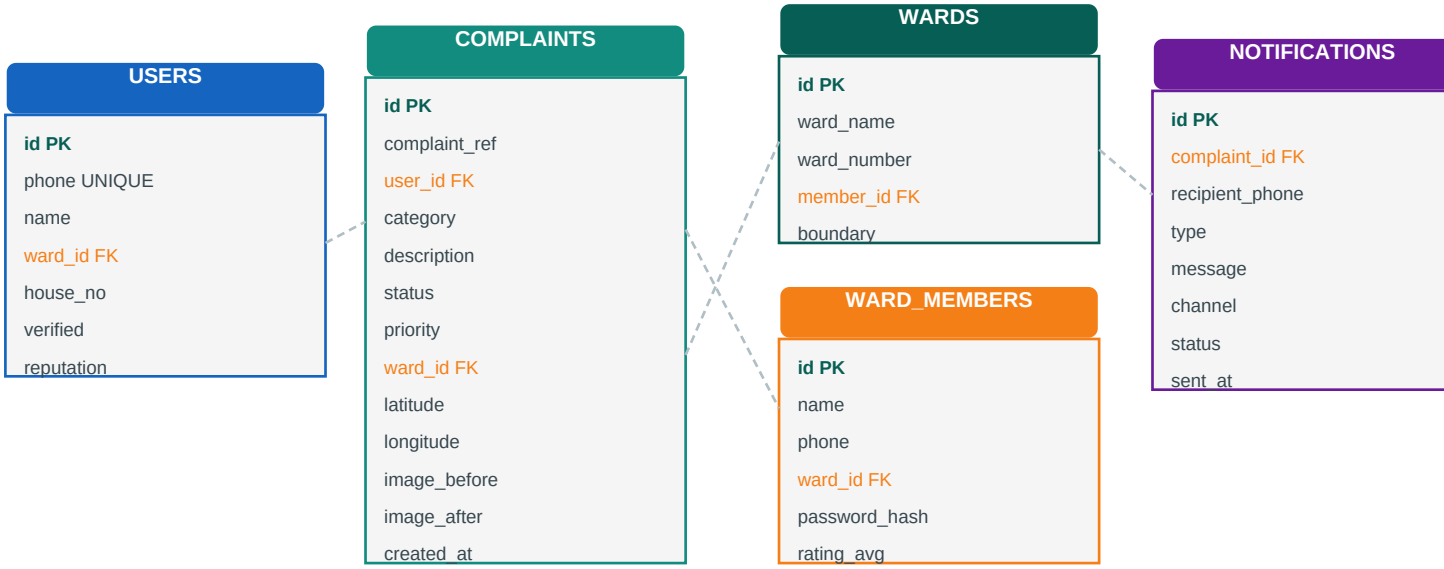


Figure 3 — Entity Relationship Diagram | PostgreSQL Database Schema

Table: users

Column	Type / Constraint
`id`	SERIAL PK
`phone_number`	VARCHAR(15) UNIQUE
`name`	VARCHAR(100)
`house_no`	VARCHAR(100)
`ward_id`	INT FK
`verified`	BOOLEAN DEFAULT FALSE
`complaint_count`	INT DEFAULT 0
`reputation_score`	FLOAT DEFAULT 5.0
`created_at`	TIMESTAMP
`last_active`	TIMESTAMP

Table: complaints

Column	Type / Constraint
`id`	SERIAL PK
`complaint_ref`	VARCHAR(20) UNIQUE
`user_id`	INT FK → users
`ward_id`	INT FK → wards
`category`	VARCHAR(50)

sub_category	VARCHAR(100)
description	TEXT
status	ENUM(pending,in_progress,resolved,closed)
priority	ENUM(low,medium,high,urgent)
assigned_to	INT FK → ward_members
image_before_url	TEXT
image_after_url	TEXT
latitude	FLOAT
longitude	FLOAT
ai_confidence	FLOAT
is_duplicate	BOOLEAN DEFAULT FALSE
master_complaint_id	INT FK
created_at	TIMESTAMP
resolved_at	TIMESTAMP

Table: wards

Column	Type / Constraint
id	SERIAL PK
ward_name	VARCHAR(100)
ward_number	INT
member_id	INT FK → ward_members
boundary_polygon	JSONB
constituency_id	INT FK

Table: ward_members

Column	Type / Constraint
id	SERIAL PK
name	VARCHAR(100)
phone	VARCHAR(15)
ward_id	INT FK
role	VARCHAR(50)
password_hash	TEXT
rating_avg	FLOAT DEFAULT 0
total_resolved	INT DEFAULT 0
is_active	BOOLEAN DEFAULT TRUE

Table: notifications

Column	Type / Constraint

id	SERIAL PK
complaint_id	INT FK
recipient_phone	VARCHAR(15)
recipient_type	ENUM(citizen,ward_member,mla)
message	TEXT
channel	ENUM(whatsapp,sms,email)
status	ENUM(sent,delivered,failed)
sent_at	TIMESTAMP

Table: ratings

Column	Type / Constraint
id	SERIAL PK
complaint_id	INT FK UNIQUE
user_id	INT FK
ward_member_id	INT FK
score	INT CHECK(1-5)
feedback	TEXT
created_at	TIMESTAMP

Table: broadcasts

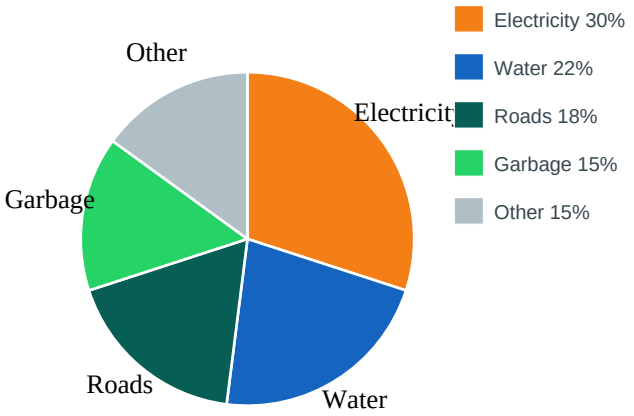
Column	Type / Constraint
id	SERIAL PK
title	VARCHAR(200)
message	TEXT
target_ward	INT FK (NULL=all)
sent_by	INT FK → ward_members
recipient_count	INT
sent_at	TIMESTAMP

§14. TECHNOLOGY STACK — COMPLETE DETAIL

Layer	Technology	Purpose	Estimated Cost
WhatsApp API	Meta WhatsApp Business API	Core messaging interface — citizens send complaints	Free 1K conversations/mo then pay-per-use
Alt. API	Twilio WhatsApp API	Easier dev setup, production fallback	$0.005 / message
Backend	Node.js + Express.js	REST API, business logic, webhooks	Free / Open Source
AI Service	Python + FastAPI	NLP microservice, ML models API	Free / Open Source
NLP + AI	spaCy + OpenAI GPT-4o	Classification, suggestions, assistant	GPT-4o ~$0.002/req
Voice → Text	Google Cloud Speech API	Convert voice notes to text (Telugu/English)	$0.004 / 15 sec
Database	PostgreSQL on Supabase	All persistent data storage	Free tier: 500MB
Cache / Session	Redis via Upstash	Chat memory, OTP storage, rate-limiting	Free tier available
File Storage	Cloudinary	Complaint photos, before/after proof	Free: 25GB
Frontend	React.js + Next.js	MLA and ward dashboards	Free / Open Source
UI System	Tailwind CSS + shadcn/ui	Clean, fast, responsive dashboard UI	Free / Open Source
Maps	Google Maps JS API	Geographic heatmaps, location display	$7 / 1,000 requests
Charts	Recharts / Chart.js	Analytics visualizations in dashboard	Free / Open Source
SMS OTP	MSG91 / Fast2SMS	OTP delivery across India	~■0.15 / SMS
Backend Host	Render.com	Deploy Node.js API server	Free tier → $7/mo
Frontend Host	Vercel	Deploy Next.js dashboards	Free tier available
CI/CD	GitHub Actions	Auto-deploy on git push	Free
Monitoring	Sentry + Uptime Robot	Error tracking, uptime alerts	Free tiers

§15. ANALYTICS, REPORTING & VISUAL INSIGHTS

The analytics module provides real-time governance insights. Sample data visualizations are shown below — these appear live in the MLA dashboard:



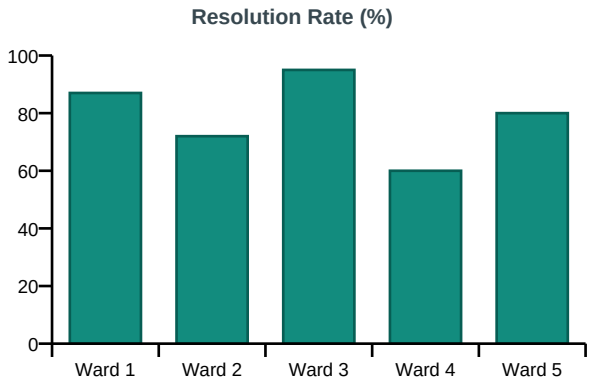


Figure 4 (left): Issue Category Distribution | Figure 5 (right): Ward-wise Resolution Rate (%)

■ KEY METRICS TRACKED: Total Complaints · Resolution Rate (%) · Avg Resolution Time (hrs) · SLA Breach Rate · Ward Performance Score · Citizen Satisfaction Rating · Peak Complaint Hours · Monthly Growth Trend · Category Breakdown · Geographic Hotspot Index

§16. PHASED DEVELOPMENT ROADMAP

PHASE 1 — MVP

Timeline: Weeks 1–6

- WhatsApp chatbot — basic conversational flow (Telugu + English)
- SMS OTP verification via MSG91
- Complaint registration with unique ID (RJNRM-YYYY-XXXX)
- PostgreSQL database setup on Supabase
- Auto-assignment to ward member by ward selection
- WhatsApp notifications — ward member + citizen confirmation
- Basic MLA web dashboard (React.js): list, filter, status update
- Three complaint statuses: Pending / In Progress / Resolved

PHASE 2 — INTELLIGENCE

Timeline: Weeks 7–10

- AI issue classification engine (GPT-4o + spaCy integration)
- Voice-to-text for WhatsApp voice notes (Google Speech API)
- Duplicate complaint detection (cosine similarity + GPS)
- Priority scoring engine (High / Medium / Low / Urgent)
- GPS location tagging + map view in dashboard
- Before/after photo upload system (Cloudinary)
- Citizen feedback & star rating system
- SLA tracking with WhatsApp auto-escalation alerts

PHASE 3 — ANALYTICS

Timeline: Weeks 11–14

- Geographic heatmaps (Google Maps API integration)
- Advanced analytics dashboard (charts, graphs, trends)
- Ward member leaderboard with performance scoring
- MLA broadcast system (WhatsApp mass messaging)
- Conversational memory for returning users
- Public transparency portal on MLA website
- Report export — PDF and Excel formats
- Emergency priority mode with Twilio Voice call alert

PHASE 4 — SCALE

Timeline: Weeks 15+

- React Native mobile app (iOS + Android)
- IVR call system for rural non-smartphone users

- Department API integrations (APEPDCL, Water Board, Municipality)
- Predictive governance AI (Facebook Prophet ML model)
- DigiLocker / Aadhaar eKYC verification integration
- Multi-constituency SaaS architecture (constituency_id)
- Telegram bot parallel channel
- AI assistant chatbot for MLA natural language queries

§17. COMMERCIAL PROPOSAL & SCALABILITY VISION

Aashvee Tech Solutions proposes a **milestone-based development engagement** for the Rajanagaram constituency implementation. Upon successful delivery, the same platform can be packaged as a **SaaS product** sold to MLAs, municipalities, and gram panchayats across Andhra Pradesh and all of India.

Engagement Model:

Milestone	Deliverable	Payment Terms
Milestone 1 (Week 2)	Requirements finalization, DB schema, API design documents	20% on signing
Milestone 2 (Week 6)	Phase 1 MVP — live WhatsApp bot + basic dashboard + notification	30% on delivery
Milestone 3 (Week 10)	Phase 2 — AI classification, proof system, ratings, SLA, escalation	25% on delivery
Milestone 4 (Week 14)	Phase 3 — Analytics, heatmaps, broadcast, MLA portal, exports	20% on delivery
Ongoing	12-month support, updates, hosting management, feature requests	5% monthly retainer

SaaS Scalability Market Opportunity:

Market Segment	Potential Customers	Revenue Model
AP State Assembly MLAs	175 MLAs in Andhra Pradesh	■8,000–■15,000/month per MLA subscription
Gram Panchayats	13,000+ panchayats in AP	■500–■2,000/month tiered subscription
Urban Local Bodies	110+ municipalities in AP	■10,000–■50,000/month enterprise plan
National SaaS Scale	4,000+ MLAs across India	■5 Crore+ Annual Recurring Revenue potential

■ STARTUP VISION: Rajanagaram is the proof-of-concept. Once validated, Aashvee Tech Solutions will package this as a multi-tenant SaaS platform — 'GovConnect by Aashvee' — enabling any constituency in India to deploy in under 48 hours. Visit: www.aashveetech.com

§18. CONCLUSION & NEXT STEPS

The **Rajanagaram Smart Governance System** is not merely a complaint management tool — it is a **data-driven digital governance platform** that transforms how an MLA's office connects with, serves, and builds trust with its constituency. By meeting citizens on WhatsApp — where they already are — it eliminates all friction from issue reporting while creating unprecedented transparency and measurable accountability.

Aashvee Tech Solutions Pvt Ltd brings the technical expertise, product thinking, and commitment to deliver this platform on time and to production-grade quality. We invite the MLA's office to partner with us in building a governance model that can become a benchmark for the entire state of Andhra Pradesh.

■ For Citizens	■ For Ward Members	■ For MLA Office
Easy WhatsApp reporting	Clear task management	Data-driven governance
No app download required	Performance visibility	Ward accountability metrics
Telugu & English support	Photo proof system	Live escalation alerts
Real-time status updates	Leaderboard recognition	Export-ready reports
Complaint history memory	Mobile-friendly dashboard	Public trust & transparency

For Aashvee Tech Solutions Pvt Ltd

Authorised Signatory
www.aashveetech.com
Date: April 2026

For MLA Office, Rajanagaram Constituency

Honourable MLA / Authorised Representative
East Godavari District, AP
Date: _____

Smart Governance for a Better Tomorrow ■

Aashvee Tech Solutions Pvt Ltd · www.aashveetech.com · Rajanagaram, Andhra Pradesh